

Notes: Alchian and Demsetz (AER, 1972)

Production, Information Costs, and Economic Organization

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1 Big picture

- **Question.** Why do firms exist, and what distinguishes the classical capitalist firm from ordinary market exchange?
- **Setting.** Several separately owned inputs cooperate in *team production*, where total output is joint and nonseparable, so individual marginal products are hard to observe directly.
- **Core challenge.** If team output cannot be cleanly decomposed into each member's contribution, then rewards cannot be tightly linked to marginal productivity. This creates *shirking* and weakens the incentive effects of decentralized market exchange.
- **Main conceptual contribution.** The firm is not defined by fiat, authority, or superior disciplinary power. Instead, the firm is a *centralized contractual arrangement* created to meter productivity and reduce shirking in team production.
- **Team-production idea.** The paper defines team production as production that uses several resources and yields an output that is *not* the sum of separable outputs. Team production is worthwhile only when its extra output exceeds the costs of organizing and policing the team.
- **Shirking result.** Under equal sharing, each team member bears only a fraction of the output loss caused by his own shirking. Private incentives therefore diverge from social incentives.
- **Main organizational result.** Efficient team production calls for a **specialized monitor** who observes input behavior and a **residual claimant** who internalizes the gain from reduced shirking. The paper's key result is that these roles should be co-located.
- **Definition of the classical firm.** The classical capitalist firm is a contractual structure with six linked elements: joint input production, several input owners, one common contractual party, unilateral rights to revise individual contracts, residual-claimant status, and the right to sell that central contractual position.
- **Comparative organizational predictions.** Profit sharing is more viable in small professional or artistic teams; corporations attenuate but partly restore the classical firm through transferability of shares and control contests; nonprofits and mutuals should exhibit more shirking; unions can be understood as monitors of hard-to-observe employer performance.
- **Input-ownership prediction.** Inputs whose user cost or depreciation is hard to observe when rented will tend to be owner-used, while resalable capital that can back the residual claimant position is more likely to be committed to the firm.
- **Broader contribution.** The firm is also a specialized private market for collecting, collating, and selling information about heterogeneous inputs and about which inputs work well together.
- **Economic takeaway.** The firm exists not because employers possess special authority, but because central monitoring plus residual claimancy economizes on information costs when production is team-based and marginal products are costly to meter.

2 Data and measurement (key objects)

2.1 Why this is a theory paper

- This is not an empirical paper. There is no sample construction, no regression table, no instrument, and no causal identification design.
- The paper’s core objects are organizational and contractual: team output, marginal-product metering, shirking, monitoring, residual claimancy, and ownership of specific inputs.
- The relevant “measurement” problem is conceptual: how can a production system measure each input’s contribution well enough to align rewards with productivity?

Interpretation. The paper studies the *information technology of organization*, not a treatment effect in data.

2.2 Team production

A compact way to summarize the paper’s production environment is

$$Z = Z(X_1, \dots, X_n), \quad (1)$$

with at least two inputs, and with output *nonseparable* in the sense that it cannot be written as

$$Z = \sum_{i=1}^n Z_i(X_i). \quad (2)$$

The paper informally characterizes this by requiring interaction among inputs, which can be summarized by a nonzero cross effect such as

$$\frac{\partial^2 Z}{\partial X_i \partial X_j} \neq 0 \quad \text{for some } i \neq j. \quad (3)$$

Team production is worth using only when the output gain from joint production exceeds the organizational cost of running the team:

$$Z(X_1, \dots, X_n) - \sum_{i=1}^n Z_i(X_i) > C_{\text{org}}. \quad (4)$$

Interpretation. The firm problem arises only when team production is productive *and* when the informational burden of running the team is nontrivial.

2.3 Multiple owners

The paper adds a third condition beyond nonseparable production: not all inputs belong to one person. That is, the relevant organizational problem is not simply joint production, but joint production among *different owners*.

Interpretation. If one person owned all inputs, shirking and metering would look different. The paper’s problem is specifically about how separately owned inputs cooperate.

2.4 Shirking and weak incentive alignment

Let ΔZ_i denote the reduction in total team output caused by shirking by member i . If member i receives only a share s_i of team output, then his private cost of shirking is only

$$s_i \Delta Z_i, \tag{5}$$

whereas the social cost is

$$\Delta Z_i. \tag{6}$$

Under equal sharing in an n -person team, $s_i = 1/n$, so each member internalizes only a fraction of the damage from his own shirking.

Interpretation. This is the paper's basic agency problem. Team members share the benefits of their effort with others, so each has too little incentive to work hard or use inputs carefully.

2.5 Monitoring

The paper uses *monitoring* broadly. It includes:

- measuring output performance,
- observing input behavior,
- apportioning rewards,
- assigning tasks or revising duties,
- and terminating or revising contracts when needed.

Interpretation. Monitoring is not just watching people. It is the whole package of activities needed to connect team output to individual incentives.

2.6 Residual claimant

If the monitor receives the residual return after other inputs are paid their contractual amounts, a convenient summary is

$$R = Z - \sum_{i \neq m} w_i, \tag{7}$$

where w_i is the promised payment to nonmonitor inputs.

Interpretation. Residual claimancy gives the monitor a direct payoff from reducing shirking and raising team output. This is why the paper wants the monitor and residual claimant to be the same person or group.

2.7 The six defining elements of the classical firm

The paper's definition of the classical firm can be summarized as follows:

1. joint input production,

2. several input owners,
3. one party common to all contracts,
4. the right to renegotiate one input's contract without reopening all other contracts,
5. residual-claimant status,
6. and the right to sell this central contractual position.

Interpretation. The firm is a bundle of contractual rights assembled to solve a metering problem. It is not a mystical zone of command.

2.8 Organizational variants studied in the paper

The paper discusses several organizational forms as variants or contrasts to the classical firm:

- profit-sharing teams,
- corporations,
- nonprofits and mutuals,
- partnerships,
- and employee unions.

Interpretation. These are not side remarks. They are comparative predictions of the theory about where monitoring problems are likely to be severe or mild.

2.9 Input ownership and user cost

The paper distinguishes between an input's productive contribution and the *user cost* or depreciation imposed on it during use. If user cost is easier to detect when the owner personally observes use, owner-use becomes relatively attractive.

Interpretation. This is why worker-owned portable tools, rented capital, and firm-owned long-lived equipment do not all look the same in equilibrium.

2.10 The firm as an information market

The monitor-employer acquires information about the talents of many heterogeneous inputs and about which combinations of inputs are productive in team settings.

Interpretation. The firm does not only *discipline*. It also *matches* and *reassigns* inputs more efficiently than purely spot-market search in some environments.

3 Models and empirical specifications (all equations + interpretation)

3.1 The nonseparable team-production problem

The paper's starting point is a production problem in which output is joint and nonseparable:

$$Z = Z(X_1, \dots, X_n), \quad Z \neq \sum_{i=1}^n Z_i(X_i). \quad (8)$$

The team arrangement is efficient ex ante if the output gain from cooperation exceeds the cost of organizing the team:

$$Z(X_1, \dots, X_n) - \sum_{i=1}^n Z_i(X_i) > C_{\text{org}}. \quad (9)$$

Interpretation. Team production creates a surplus, but that surplus is endangered by the difficulty of attributing who produced what.

3.2 Why ordinary market metering breaks down

In ordinary market exchange with separable outputs, the seller's receipt is tightly linked to his own output. In team production, total output is observed only jointly. This means the mapping

$$\text{reward}_i \longleftrightarrow \text{marginal product}_i \quad (10)$$

is costly to implement.

A useful schematic way to summarize the distortion is

$$\frac{\partial \text{reward}_i}{\partial X_i} < \frac{\partial Z}{\partial X_i} \quad \text{under decentralized equal sharing.} \quad (11)$$

Interpretation. The team's total surplus may be observable, but each member's contribution is not. That wedge is the source of shirking.

3.3 Shirking under equal profit sharing

Suppose input owner i bears only share s_i of the consequences of his own reduced effort or careless behavior. Then private and social costs diverge:

$$\text{Private cost of shirking}_i = s_i \Delta Z_i, \quad \text{Social cost of shirking}_i = \Delta Z_i. \quad (12)$$

With equal sharing, $s_i = 1/n$, so a larger team typically means a weaker private penalty for shirking.

Interpretation. This gives the paper's prediction that pure profit-sharing arrangements become less attractive as team size rises, unless other monitoring mechanisms compensate.

3.4 Specialized monitoring as the first solution

One way to reduce shirking is to appoint a specialist monitor who devotes effort to observing behavior and measuring performance. But then a second-order problem appears: who monitors the monitor?

Interpretation. The firm's contract must solve both the workers' shirking problem and the monitor's shirking problem.

3.5 Why the monitor should be the residual claimant

The paper's core solution is to combine monitoring with residual claimancy:

$$R = Z - \sum_{i \neq m} w_i. \quad (13)$$

Because the monitor captures the residual, he gains directly from reducing shirking and increasing team output.

Interpretation. Residual claimancy is not mainly compensation for “bearing uncertainty” in the Knightian sense. In this paper it is primarily an incentive device for monitoring.

3.6 Why the monitor must also control contract revision

Residual claimancy alone is not enough. The central monitor must also be able to revise individual contracts, change team membership, and terminate underperforming members without reopening every agreement simultaneously.

The paper's contractual logic can be summarized as:

$$\text{team efficiency} \Rightarrow \text{monitoring} \Rightarrow \text{bilateral contracts with one central party} \Rightarrow \text{classical firm}. \quad (14)$$

Interpretation. The firm is a governance structure that economizes on repeated multilateral renegotiation in the presence of costly performance measurement.

3.7 The classical firm as a bundle of rights

The paper's explicit list of firm-defining conditions can be written schematically as

$$\mathcal{F} = \{\text{team production, multiple owners, central contracting party, independent contract revision, residual claim,} \quad (15)$$

Interpretation. All six parts matter. Remove the residual claim and monitor incentives weaken; remove independent revision and discipline becomes cumbersome; remove alienability and the monitor's position becomes less contestable.

3.8 Why authority is not the primitive

A major conceptual move in the paper is to deny that the firm is essentially a zone of command or fiat. Employer and employee both remain parties to voluntary contracts and either side can terminate the relationship.

Interpretation. What looks like “authority” is better understood as the monitor's contractual right to propose new terms or to terminate and replace team members.

3.9 Comparative organizational forms

3.9.1 Profit-sharing firms

Profit sharing becomes relatively more viable when team size is small and when reciprocal monitoring is effective. The paper especially emphasizes small professional or artistic teams, where centralized close monitoring is expensive.

Interpretation. The theory does not say profit sharing never works. It says it works best where the monitor's own job is hard to discipline and where team size is limited.

3.9.2 Corporations

Corporations diffuse ownership for financing reasons, which weakens direct stockholder monitoring. But transferability of shares, proxy fights, and control contests can reconcentrate residual claims temporarily and help discipline managers.

Interpretation. The corporation is a modified and attenuated version of the classical firm, with managerial monitoring constrained by capital-market governance rather than by direct day-to-day owner oversight.

3.9.3 Nonprofits and mutuals

Where gains from improved management cannot be capitalized into tradable claims, competition by outside monitors becomes weaker.

Interpretation. The paper predicts more managerial slack or shirking in nonprofit and mutual forms, especially in activities where market reorientation is less valuable or less desired.

3.9.4 Partnerships

Partnerships are most natural in small teams of professional or artistic inputs where members can monitor each other and where specialized management is hard to perform from the outside.

Interpretation. Lawyers, doctors, and similar professionals are not anomalies. They are exactly the cases where the theory predicts partnership-like forms.

3.9.5 Employee unions

Unions are interpreted as monitors hired by employees to police employer performance on dimensions that are hard for each worker to verify individually, such as contingent fringe benefits.

Interpretation. Monitoring problems run in both directions: employers monitor employees, but employees may also need monitors of employers.

3.10 Team spirit as a low-cost policing device

The paper also discusses team spirit and loyalty as informal ways to reduce shirking. If each team member internalized the common interest in nonshirking, the organization would be more efficient even without stronger formal policing.

Interpretation. “Culture” matters here only because it reduces incentive distortions. Team spirit is economically valuable insofar as it substitutes for costly formal monitoring.

3.11 Which inputs will the firm own?

The paper gives two broad reasons why some inputs are more likely to be owned by the firm or by the residual claimant group.

1. **Collateral and loss-bearing.** The residual claimant must be able to cover negative residuals, so resalable capital assets are natural candidates for firm ownership.
2. **User-cost monitoring.** Inputs whose depreciation or abuse is hard to detect when rented tend to be owner-used rather than rented out.

A compact way to summarize the user-cost logic is

$$\text{own rather than rent when } C_{\text{detect user cost} \mid \text{rental}} > C_{\text{detect user cost} \mid \text{owner use}}. \quad (16)$$

Interpretation. This is why workers often own portable tools, while long-lived, pledgeable capital is more likely to be committed to the firm.

3.12 The firm as a specialized private market

Because the employer-monitor repeatedly observes many heterogeneous inputs, he acquires better information about their talents and about productive combinations. This makes internal reassignment, promotion, and regrouping of resources potentially cheaper than fresh market search.

Interpretation. The firm is not merely a disciplinary hierarchy. It is also an information broker and matching institution.

4 Identification and instruments (what makes the theoretical results credible)

4.1 No empirical identification or instruments

- The paper does not estimate a coefficient from data.
- There is no instrument, no natural experiment, no panel design, and no formal structural estimation.
- The relevant credibility question is therefore: which assumptions are sufficient to generate the classical firm as an efficient contractual solution?

Interpretation. This is a theory paper. Its force comes from internal logic, not from statistical identification.

4.2 The assumptions that do the work

The central argument relies on five key premises:

1. team production is genuinely nonseparable,
2. several separately owned inputs cooperate,
3. individual marginal products are costly to meter,
4. shirking is easier when reward and productivity are weakly linked,
5. and monitoring can be improved by centralizing residual claimancy and contract revision.

Interpretation. If any of these premises is removed, the classical firm is less obviously the right organizational form.

4.3 Why the paper rejects “flat” explanations

A major identification step in the paper is conceptual: it argues that firms do not differ from markets because employers possess some intrinsically stronger power of command. The employer can reward, renegotiate, or terminate, but these are still contractual responses.

Interpretation. The theory identifies the firm’s essence in *monitoring and contractual centralization*, not in authority per se.

4.4 How this differs from Coase

The paper explicitly positions itself relative to Coase.

- Coase emphasizes the cost of using markets to transact.
- Alchian and Demsetz emphasize the cost of *measuring individual performance inside team production*.
- The paper argues that lower market-transaction costs do not always reduce the role of the firm; some technologies lower contracting costs while simultaneously making team production more important and marginal-product measurement harder.

Interpretation. This is why the paper uses the factory system example: centralized power sources could reduce some market costs and yet still expand the domain of the firm because monitoring team use of equipment became more important.

4.5 How this differs from Knight

The paper also distances itself from Knight-style uncertainty-bearing theories.

- Knight links entrepreneurship to bearing uncertainty.
- Alchian and Demsetz argue that residual claimancy is mainly valuable here because it disciplines the monitor.

- Risk-bearing may matter for who finances the firm, but it is not the core reason the classical firm exists.

Interpretation. Residual claims matter here because they sharpen policing incentives, not because the owner is uniquely willing to bear risk.

4.6 Testable implications generated by the theory

Even without formal econometrics, the paper generates several empirical-style predictions.

- Profit sharing should be concentrated in small teams and especially in professional or artistic settings.
- Corporations should need governance mechanisms that reconcentrate control when management underperforms.
- Nonprofits and mutuals should exhibit more slack because managerial improvements are not capitalized into transferable residual claims.
- Unions should arise where employer performance is hard for individual workers to meter.
- Owner-use should be more common when an input's user cost is hard to observe under rental.

Interpretation. The paper is not nonfalsifiable. It proposes an organizational map that can be checked against patterns of ownership and governance.

4.7 What the paper does not fully model

- It does not build a fully explicit mathematical optimization problem for all organizational forms.
- It does not formally derive the initial distribution of ownership rights or why one person does not simply own all inputs.
- It does not estimate the magnitude of the monitoring-cost savings.
- It does not explain every institution commonly called a firm, especially conglomerates and holding companies, with equal force.

Interpretation. The paper is best read as a sharp conceptual theory of one important class of firms rather than a universal theory of all business organizations.

4.8 Why the argument remains important

The paper's durability comes from a clear reversal of the usual causal order. Instead of assuming that marginal products are somehow obvious and then inferring payment rules, the paper argues that the *reward system itself* shapes productivity by shaping incentives and monitoring.

Interpretation. Organization is not a passive wrapper around production. It is part of the production technology whenever information is costly.

5 Tables and figures

This paper has no conventional empirical tables or data figures. The most useful way to read its evidence is through summary exhibits that collect the paper’s theoretical predictions and conceptual distinctions.

5.1 Exhibit 1: The metering problem

Table 1: How the paper moves from team production to the firm

Step	Content	Economic message
Team production	Output is joint and nonseparable across multiple inputs.	Total surplus is larger than under separate production, but attribution of contribution becomes difficult.
Costly metering	Individual marginal products are hard to observe directly from total output.	Market rewards no longer map cleanly into productivity.
Shirking	Each member bears only part of the output loss caused by his own reduced effort.	Incentives are diluted inside the team.
Monitoring	A specialist can observe performance and reassign tasks.	Organization can reduce agency losses, but the monitor himself must be disciplined.
Residual claimancy	Give the monitor the residual return.	The monitor internalizes the gain from reducing shirking.
Central contracting	Let the monitor revise and terminate individual contracts.	The classical firm emerges as a low-cost policing device.

Read. The logic is sequential: team production creates surplus, costly metering creates shirking, and central monitoring plus residual claimancy creates the firm.

Economic message. The paper is not “firms versus markets” in general. It is specifically about *team production plus information costs*.

5.2 Exhibit 2: The six defining features of the classical firm

Table 2: What the paper means by the classical firm

Feature	Definition	Why it matters
Joint production	Inputs are used together in a team.	Creates scope for gains from cooperation.
Several owners	Inputs belong to different people.	Makes incentive alignment and contracting nontrivial.
One common party	One agent contracts with all other input owners.	Centralizes governance and reduces multilateral bargaining costs.
Independent revision	One input's contract can be changed without reopening every other contract.	Makes discipline and reassignment feasible.
Residual claim	The central agent receives the residual return.	Aligns monitoring incentives with team output.
Alienability	The central contractual position can be sold.	Makes the monitor's role contestable and transferable.

Read. The firm is a linked bundle of governance rights, not just a location where people work together.

Economic message. Residual claimancy alone is not enough; it must be paired with central contracting rights.

5.3 Exhibit 3: Predicted organizational forms

Table 3: How the theory maps into different organizations

Organization	When it is likely to work best	Main strength in the paper	Main weakness in the paper
Profit-sharing team	Small teams; stronger self-policing; limited central-management advantage.	Shares incentives broadly across members.	As team size grows, each member bears too little of the loss from shirking.
Partnership	Small professional or artistic teams.	Reciprocal monitoring works and close centralized supervision is difficult anyway.	Hard to scale when many members must share the residual.
Corporation	Large-scale financing with many outside investors.	Transferable shares and control contests can discipline management.	Diffuse ownership weakens direct monitoring and day-to-day residual incentives.
Nonprofit or mutual	Settings where market redirection is less valued or less desired.	Can serve missions that do not fit tradable residual claims well.	Managerial improvements are not capitalized into tradable claims, so slack is more likely.
Union	Employment relations with hard-to-monitor employer performance, especially fringe benefits.	Acts as a monitor for workers.	Adds another layer of organization and bargaining cost.

Read. The paper is a comparative theory of organizational form, not only a theory of the proprietary firm.

Economic message. Different organizations appear when the monitoring problem takes different shapes.

5.4 Exhibit 4: Who owns which inputs?

Table 4: Predictions about ownership of inputs

Input characteristic	Likely ownership pattern	Reason in the paper
Resalable capital with long service life	More likely committed to the firm or residual claimant.	It can back negative residuals and makes the monitor's promises more credible.
Portable tools whose user cost is easy for the worker to observe while using them	More likely worker-owned.	The user-owner can police depreciation or abuse more cheaply than an absentee owner.
Human capital or labor services	Usually not firm-owned.	Human wealth is hard to collateralize and self-monitoring by the worker is relatively direct.
Assets whose user cost is hard to detect after rental use	More likely owner-used than rented.	Rental requires a premium because expected abuse is harder to police.

Read. Ownership patterns are shaped by information costs, not only by production technology.

Economic message. The boundary of the firm is partly a boundary of who can best monitor an asset's deterioration and use.

5.5 Exhibit 5: Comparison with Coase and Knight

Table 5: How this paper differs from two classic alternatives

Reference point	What that view emphasizes	What Alchian–Demsetz add or change
Coase	Costs of using the market to transact.	The crucial cost here is not just writing contracts across markets, but measuring individual contributions inside team production.
Knight	Entrepreneurship as reward for bearing uncertainty.	Residual claimancy matters mainly because it disciplines the monitor and reduces shirking, not primarily because the owner bears risk.
Traditional authority view	Firm as command or fiat.	The firm is instead a central contractual agent with rights to renegotiate and replace inputs.

Read. The paper is an intervention in the theory-of-the-firm debate.

Economic message. Its novelty lies in making costly measurement and policing the centerpiece of organizational design.

6 Short summary

- The paper explains the firm as a response to costly metering of marginal products in team production.
- Team production is valuable because joint use of inputs can create more output than separate production, but it also makes shirking hard to detect.
- Equal sharing dilutes incentives because each member bears only part of the output loss caused by his own shirking.
- The efficient response is a specialized monitor who is also the residual claimant and who can revise bilateral contracts with team members.
- The classical firm is therefore a contractual bundle, not a zone of fiat or intrinsic authority.
- The theory predicts where partnerships, corporations, nonprofits, unions, and different ownership patterns of inputs should be observed.

7 Conclusion

7.1 Core conclusion

The paper's core conclusion is that the classical capitalist firm is an efficient contractual response to the problem of monitoring team production when individual marginal products are costly to measure. The firm's essence is not authority, but the co-location of monitoring, residual claimancy, and rights to revise individual contracts.

7.2 Economic intuition

When production is team-based, each individual can hide inside joint output and let others bear most of the cost of his shirking. A monitor can reduce this problem, but only if the monitor himself has a strong incentive to do the job well. Giving the monitor the residual return and central contractual status creates that incentive.

7.3 Takeaway

This paper remains foundational because it turns organization into an information problem. Firms arise not merely because markets are costly, and not mainly because someone bears uncertainty, but because team production requires a low-cost way to meter performance, police shirking, and assemble productive combinations of heterogeneous inputs.